
Position: Learned Aleatoric Uncertainty Should Never Be Reported Without a Measured Noise Floor

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Abstract

Predictive uncertainty is routinely split into an epistemic part, read as ignorance that more data can remove, and an aleatoric part, read as noise that nothing can remove. The identity behind that split is exact. The reading is not. This position paper argues that a learned aleatoric estimate is the true floor plus the model’s own error, of a magnitude and a sign that cannot be known without measuring the floor, and that the community should stop reporting it as evidence of irreducibility: report it against a separately measured floor wherever one can be obtained, with the signed gap, and withdraw the claim of irreducibility where it cannot. For regression with a learned noise scale, the gap at the population optimum is the sum of two non-negative model-error terms, the ensemble’s mean-fit error and the misspecification of the noise model, which extends a known result on bias contamination; fitting a finite sample supplies a third term that deflates it. For classification, a sign analysis shows a divergence term that inflates the reported floor and a sharpness term that deflates it, and the two cancel, so agreement with a measured floor is not evidence that the estimate is right. Subtracting the epistemic term from the gap isolates the ensemble’s bias, the part of excess risk that no term of the decomposition reports. On CIFAR-10H, against 511,000 human judgements, the signed gap crosses zero as capacity grows, earliest on the most ambiguous inputs, while the epistemic term is non-monotone in capacity and never warns; the finding survives deconvolving annotator error at ten times the rate the data permit. A pre-registered survey of 50 recent papers finds that of the 38 reporting such an estimate, 5 compare it to any real floor and 22 to nothing. We address the strongest objection, that floors are unmeasurable in most real tasks, by conceding it and narrowing the claim: irreducibility is an empirical

claim, and a claim without evidence should be reported as one.

1. Introduction

A practitioner with a model that will not improve faces one question: is the remaining error a fact about the world, or a fact about the model? The standard instrument for answering it is the decomposition of predictive uncertainty into epistemic and aleatoric parts (Depeweg et al., 2018; Hüllermeier & Waegeman, 2021). The epistemic part is read as ignorance about which function is right, removable by labelling more data; it is what active-learning acquisition functions rank candidates by (Houlsby et al., 2011) and what drives exploration in reinforcement learning (Osband et al., 2016). The aleatoric part is read as ambiguity in the label given the features, and it is the term that ends a project: if the noise floor has been reached, there is nothing left to do.

The identity behind the split is exact. It is an algebraic consequence of the definition of mutual information and it holds for any posterior whatsoever, well-specified or not, converged or not. What is *not* exact is the reading of the two terms, and in particular the reading of the aleatoric term as irreducible. That reading requires the hypothesis class to contain the truth. When it does not, the structure the class cannot represent still has to be accounted for somewhere, and in a model with a learned noise scale the only free parameter available is the noise scale itself. A linear model with a heteroscedastic Gaussian head fitted to $y = \sin(x) + \varepsilon$ reports large aleatoric uncertainty and near-zero epistemic uncertainty: its members agree, and they agree on a flat line. The system’s claim is that predictions here are noisy by nature. One extra basis function would remove the error entirely.

Position. A learned aleatoric estimate is not a measurement of the noise floor. It is the floor plus the model’s own error, of unknown magnitude and, in classification, of unknown sign. The machine learning community should stop reporting it as evidence of irreducibility. Wherever repeated labels, replicate measurements, or human-agreement data can be obtained, learned

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aleatoric uncertainty should be reported against that separately measured floor, together with the signed gap between them. Where no floor can be measured, the claim that the uncertainty is irreducible should be withdrawn rather than assumed.

What is new is not the caveat but its sign, and a measurement of it. That the decomposition is model-relative is in the literature (Hüllermeier & Waegeman, 2021; Der Kiureghian & Ditlevsen, 2009), and Jiménez et al. (2025) have argued, in a position paper, that unaccounted bias contaminates second-order estimates by inflating the aleatoric term and deflating the epistemic one. We take that as established and build on it. What it leaves open is the direction of the error once the sample is finite and the head is a softmax, and whether the gap can be measured rather than argued. Section 3 shows that the one-sided story is a population statement: fitting the sample deflates the reported floor in regression, and in classification a sharpness term deflates it while a divergence term inflates it, so the sign of the error cannot be assumed in either setting and the two terms cancel. A model can report an aleatoric estimate that matches a measured floor exactly while being wrong in both directions at once. That is the case for measuring: not that the reported floor is too high, but that nothing about the estimate says which way it is wrong. The practice is also measurable, and we measured it: in a random sample of 50 recent arXiv papers, of the 38 that report a learned aleatoric estimate, 5 compare it to any real external floor and 22 compare it to nothing (Section D).

The position is falsifiable in the ordinary way. If the gap between learned aleatoric uncertainty and a directly measured floor turns out to be small, or to carry no information beyond what proper scoring rules already provide, then the reporting requirement is not worth its cost. Section 5 is designed so that this could happen, and E1 is the experiment to run first.

Why the term is invisible, not merely hard to see. Of the four parts of classical excess risk (Bottou & Bousquet, 2008), three have a counterpart in the uncertainty decomposition and approximation error does not. The reason is one line, $p(f | S, \mathcal{M}) = 0$ for every $f \notin \mathcal{M}$: a posterior has no quantity with which to say its class is the wrong shape, and exact inference has the same blind spot as approximate inference (Drafer, 1995; Grünwald & van Ommen, 2017). Section 3 develops this.

Contributions.

1. An exact decomposition of the gap between the reported aleatoric term and the true floor for regression at the population optimum (Proposition 3.1); a sign

analysis of the same gap for classification (Proposition 3.2); and the finite-sample deflation that turns the one-sided contamination result of Jiménez et al. (2025) into a two-sided one.

2. The gap *minus* the epistemic term identified with the ensemble’s squared bias plus the noise-model error (Equation (5)): the part of excess risk no term reports, in nats.
3. A law-of-total-variance account of why enriching the feature set can only lower the floor, and the observation that a biased sampling frame *lowers* epistemic uncertainty while deployment error does not fall (Section 4).
4. A measurement protocol and three experiments quantifying the gap, its sign as a function of capacity, and the epistemic term’s failure to warn (Section 5).

Scope. We do not argue that the decomposition should be abandoned, that aleatoric estimates are useless, or that every task admits a measurable floor. The last of these is the strongest objection to the position and we treat it at length in Section 6.

2. Two Decompositions

We use one set of symbols throughout. $\mathcal{D}$ is the joint distribution over inputs x and labels y , ℓ a loss, S a sample of n points drawn independently from $\mathcal{D}$, and $\mathcal{M}$ the hypothesis class, meaning the set of functions the architecture can express. A parameter θ picks one function out of $\mathcal{M}$, and f_S is the predictor a training run returns.

2.1. The error ledger

The risk $R(f)$ of a predictor is its expected loss on fresh data from $\mathcal{D}$. The Bayes risk R^* is the smallest risk any function can reach; it is the floor set by ambiguity in the problem, and the function that reaches it is the Bayes predictor f^* . The excess risk above that floor splits three ways (Bottou & Bousquet, 2008):

$$R(f_S) - R^* = \underbrace{R(f_{\mathcal{M}}) - R^*}_{\text{approximation}} + \underbrace{R(\hat{f}_S) - R(f_{\mathcal{M}})}_{\text{estimation}} + \underbrace{R(f_S) - R(\hat{f}_S)}_{\text{optimisation}}, \quad (1)$$

where $f_{\mathcal{M}}$ is the best function in the class and $\hat{f}_S$ the empirical risk minimiser. Approximation error asks how good the best function the architecture can express is; more data does not reduce it. Estimation error is the price of a finite sample, falling in n and rising with the richness of $\mathcal{M}$. Optimisation error is the optimiser returning something other than the empirical minimiser. Under squared loss the Bayes predictor is the conditional mean $m(x) = \mathbb{E}[y | x]$ and the Bayes risk is

$\mathbb{E}[\text{Var}(y \mid x)]$, the noise term of the bias–variance decomposition (Section A). Every term of Equation (1) needs labels to observe.

2.2. The uncertainty ledger

Let $p(\theta \mid S)$ be a posterior over the parameters of $\mathcal{M}$. At an input x , the entropy of the posterior predictive splits into a mutual information and an expected conditional entropy (Depeweg et al., 2018; Smith & Gal, 2018):

$$\underbrace{\mathbb{H}[\mathbb{E}_\theta p(y \mid x, \theta)]}_{\text{total}} = \underbrace{\mathbb{I}(y; \theta \mid x)}_{\text{epistemic}} + \underbrace{\mathbb{E}_\theta \mathbb{H}[p(y \mid x, \theta)]}_{\text{aleatoric}}. \quad (2)$$

In practice the posterior is replaced by M parameter settings, a deep ensemble (Lakshminarayanan et al., 2017) or dropout samples (Gal & Ghahramani, 2016), and the whole content of the identity is the order in which two operations are applied: average the members’ distributions and then take the entropy, or take each member’s entropy and then average. The difference is how much the members disagree. It cannot be negative, because entropy is concave, and it is computable at prediction time with no labels at all.

2.3. Why the second ledger is read as actionable

The two parts are read as calling for different actions: the epistemic part as ignorance about θ that data would remove, which acquisition functions rank by (Houlsby et al., 2011); the aleatoric part as ambiguity no further labels would remove, the standard name for which is *irreducible* noise; the total as the basis for abstention (Geifman & El-Yaniv, 2017). Both identities are exact. But the first says what the model gets wrong and the second what it reports about its own ignorance, and nothing makes the second track the first.

3. The Mapping and the Missing Term

3.1. Three terms match, one does not

Table 1 places the two decompositions side by side. Bayes error is what the aleatoric term is supposed to measure. Estimation error is what the epistemic term does measure: the mutual information is exactly the expected information a label at x would carry about θ , which is the quantity that shrinks as the sample pins θ down. Optimisation error appears weakly, because members that settle in different basins disagree, and that disagreement is read as epistemic. Approximation error appears nowhere.

The reason is one line. A posterior over the parameters of a class $\mathcal{M}$ places no mass on functions outside it:

$$p(f \mid S, \mathcal{M}) = 0 \quad \text{for all } f \notin \mathcal{M}. \quad (3)$$

A model therefore has no quantity with which to express that its class is the wrong shape. This is routinely conflated

Table 1. Each source of excess risk and the uncertainty term that reports it. The last row is reported by nothing.

Error source	Uncertainty term	Reported?
Bayes error	aleatoric	yes
Estimation error	epistemic	yes
Optimisation error	epistemic, partly	weakly
Approximation error	none	no

with the better-known claim that variational and dropout posteriors underestimate epistemic uncertainty; that is about inference quality and is fixed by better inference, whereas Equation (3) is about the definition of the posterior and is fixed by nothing. Exact inference in a misspecified class has the same blind spot (Draper, 1995; Grünwald & van Ommen, 2017). Outside deep learning the gap is patched by hand: Kennedy & O’Hagan (2001) add a discrepancy function $\delta(x)$ alongside the simulator precisely because it cannot be recovered from parameter uncertainty. Bayesian deep learning has no δ .

The rest of this section asks where the error goes instead, and answers it exactly for regression and exactly for classification. The two answers differ in a way that matters for what may be claimed.

3.2. Regression: the noise scale absorbs the residual

Let $m(x) = \mathbb{E}[y \mid x]$ and $s^2(x) = \text{Var}(y \mid x)$, and let each ensemble member emit a mean $\mu_\theta(x)$ and a variance $\sigma_\theta^2(x)$, with $b_\theta(x) = \mu_\theta(x) - m(x)$ its pointwise mean error. Write $u_{\text{ale}}(x) = \mathbb{E}_\theta \mathbb{H}[p(y \mid x, \theta)]$ for the learned aleatoric term, $u_*(x) = \mathbb{H}[p(y \mid x)]$ for the true floor, and $\Delta(x) = u_{\text{ale}}(x) - u_*(x)$.

Proposition 3.1. *If each member’s variance head attains the conditional negative log-likelihood optimum given its own mean function, then $\sigma_\theta^2(x) = s^2(x) + b_\theta(x)^2$ and*

$$\begin{aligned} \Delta(x) = & \underbrace{\frac{1}{2} \mathbb{E}_\theta \log \left(1 + \frac{b_\theta(x)^2}{s^2(x)} \right)}_{\text{mean-fit error}} \\ & + \underbrace{\text{KL}(p(y \mid x) \parallel \mathcal{N}(m, s^2))}_{\text{noise-model error}}. \end{aligned} \quad (4)$$

Both terms are non-negative, so $\Delta(x) \geq 0$, with equality if and only if every member’s mean equals the Bayes predictor at x and the true conditional law is Gaussian.

Proof sketch. The NLL optimum in σ_θ^2 is $\mathbb{E}[(y - \mu_\theta)^2 \mid x] = s^2 + b_\theta^2$; substitute into the Gaussian entropy and use $\frac{1}{2} \log(2\pi e s^2) - \mathbb{H}[p] = \text{KL}(p \parallel \mathcal{N}(m, s^2))$ (Section A). $\square$

Three things follow, and the first corrects a claim it is tempting to make.

The gap is not approximation error. Decomposing the mean error across the ensemble, $\mathbb{E}_\theta b_\theta(x)^2 = (\mathbb{E}_\theta \mu_\theta(x) - m(x))^2 + \text{Var}_\theta(\mu_\theta(x))$, and integrating the first term over x recovers the excess squared-loss risk of the ensemble mean, which Section 2 splits into approximation, estimation and optimisation error. So Δ reports total excess risk, not approximation error alone. It isolates approximation error only in the limit of a converged optimiser and a large sample, where the other two vanish. We state this because the weaker claim is the true one and the stronger one is what a reader of the informal argument would take away.

Some of the gap is already reported. Moment-matching the ensemble’s predictive mixture gives $u_{\text{epi}}(x) \approx \frac{1}{2} \log(1 + \text{Var}_\theta(\mu_\theta)/\mathbb{E}_\theta \sigma_\theta^2)$, so the $\text{Var}_\theta(\mu_\theta)$ part of the mean error is exactly what the epistemic term already sees. Subtracting it, and expanding to leading order in the regime where the errors are small relative to the noise,

$$\Delta(x) - u_{\text{epi}}(x) = \frac{(\mathbb{E}_\theta \mu_\theta(x) - m(x))^2}{2s^2(x)} + \text{KL}(p \parallel \mathcal{N}) + O(\cdot). \quad (5)$$

The right-hand side is the squared *bias* of the ensemble (the error common to every member, which no amount of disagreement can reveal) plus the error in the shape of the noise model. This is the missing term of Table 1, in nats, and it is the quantity E1 measures.

The bound does not need Gaussian noise. The proposition assumes the head is Gaussian, not the world: a skewed or heavy-tailed truth enlarges the second term of Equation (4) rather than invalidating it, since the Gaussian is the maximum-entropy law at fixed variance. Section 5.2 measures both terms separately.

The one-sided guarantee is a population statement. Proposition 3.1 places the variance head at the negative log-likelihood optimum *on the true distribution*. A trained network sits at the optimum on its sample, and the two differ in a direction that matters: an overparameterised head fits training residuals, which are smaller than the true noise, so σ_θ^2 can fall below s^2 and Δ below zero. Section 5.2 measures $\Delta = -0.069$ at the widest model in the sweep, in a regression problem with no softmax anywhere. So $\Delta \geq 0$ characterises the population optimum, not trained models, and the honest reading of Equation (4) is that it names the two ways a class can *inflate* the reported floor while finite-sample fitting supplies a third that deflates it.

3.3. Classification: the gap is signed

The regression argument needs a free noise-scale parameter to absorb the residual. Classification has none, because

the softmax output *is* the predictive distribution, and the conclusion changes accordingly.

Proposition 3.2. *For a member with predictive $p_\theta = p(\cdot | x, \theta)$ and true conditional $p = p(\cdot | x)$,*

$$\Delta(x) = \underbrace{\mathbb{E}_\theta \text{KL}(p \parallel p_\theta)}_{\text{divergence: inflates}} - \underbrace{\mathbb{E}_\theta C_\theta(x)}_{\text{sharpness: deflates}}, \quad (6)$$

where $C_\theta(x) = \langle p - p_\theta, -\log p_\theta \rangle = \text{CE}(p, p_\theta) - \mathbb{H}[p_\theta]$.

Proof. $\text{CE}(p, p_\theta) = \mathbb{H}[p] + \text{KL}(p \parallel p_\theta)$ by definition, and $C_\theta = \text{CE}(p, p_\theta) - \mathbb{H}[p_\theta]$ by construction, so $\mathbb{H}[p_\theta] - \mathbb{H}[p] = \text{KL}(p \parallel p_\theta) - C_\theta$. Averaging over θ gives Equation (6). $\square$

The first term is non-negative and behaves as in regression: a class that cannot resolve the structure at x returns a smoothed predictive and reports too much ambiguity. The second term is positive exactly when the member is sharper than the truth, and modern networks are systematically sharper than the truth (Guo et al., 2017). It can be large. For $p = (\frac{1}{2}, \frac{1}{2})$ and an overconfident $p_\theta = (0.99, 0.01)$, the divergence term is 1.61 nats and the sharpness term is 2.25, so the reported floor sits 0.64 nats *below* the true one.

What this proposition is and is not. Proposition 3.2 is an identity, not a decomposition in the sense of Proposition 3.1. C_θ is defined so that it holds, and both terms require p , which is what the floor supplies; it buys interpretation, not measurement. Proposition 3.1 is different in kind: the optimum assumption does real work, and E4 measures its two terms separately. What the sign analysis makes precise is that the two mechanisms moving the reported floor in classification carry opposite signs, and it is that fact, not the identity itself, that the argument below uses.

One mechanism, two settings. Taken with the previous subsection, a single statement covers both: *misspecification inflates the reported floor, and fitting the sample deflates it*. In classification the two appear as the two terms of Equation (6); in regression as Equation (4) against the finite-sample shrinkage of σ_θ^2 . Neither is a property of the world, and they carry opposite signs, so they cancel. A model can therefore report an aleatoric estimate that matches a measured floor exactly while being wrong in both directions at once. Agreement with the floor is not evidence that the estimate is right, and the sign of the error cannot be assumed in either setting. This is what makes the reporting requirement a requirement rather than a refinement: there is no regime in which u_{ale} alone can be read as a bound.

The sign is itself the diagnostic. $\Delta(x) > 0$ says the class cannot resolve the structure at x ; the remedy is capacity, features or a sharper target. $\Delta(x) < 0$ says the model claims certainty that annotators do not share; the remedy

is calibration, and this is the dangerous failure. Neither reading is available without the floor. Section 5.1 measures the sign as a function of capacity and finds that it changes.

3.4. Related work

The model-relativity of the split is stated by Hüllermeier & Waegeman (2021) and Der Kiureghian & Ditlevsen (2009), and Wimmer et al. (2023) question the measures themselves. Closest to this paper is Jiménez et al. (2025), who argue in a position paper that unaccounted bias inflates the aleatoric estimate and deflates the epistemic one, and that second-order methods capture only part of the epistemic variance. Their bias-contamination argument is the regression half of Proposition 3.1, and we treat it as established. What we add is the other direction of the error, the sign analysis for classification with the cancellation it permits, the residual identity Equation (5), and a measurement on a real benchmark. Mucsányi et al. (2024) benchmark nineteen estimators across thirteen tasks and find that none disentangles the two sources in practice, the empirical form of the same conclusion. Baan et al. (2022) argue that calibration should be assessed against the full distribution of human judgements rather than the majority label and propose instance-level measures for it; the protocol of Section 7 is that measurement applied to the aleatoric term, and their observation that annotator error is folded into the judgement distribution is the confound Section 5.1 treats. The structural term is handled by hand outside deep learning (Draper, 1995; Kennedy & O’Hagan, 2001) and its consequences for misspecified inference are known (Grünwald & van Ommen, 2017; Berk, 1966; Kleijn & van der Vaart, 2012). Seitzer et al. (2022) identify the gradient pathology that makes Proposition 3.1 bite; Bengs et al. (2022) and Meinert et al. (2023) show evidential estimators are not identified as their names suggest. Measured floors exist (Peterson et al., 2019; Beyer et al., 2020; Northcutt et al., 2021) but have not been used to audit a model’s own aleatoric estimate.

4. What the Identity Assumes

Equation (2) is unconditional: it is an algebraic consequence of the definition of mutual information and holds for any posterior whatsoever. What is conditional is the reading of the two terms, that the epistemic part is ignorance a modeller could remove and the aleatoric part is ambiguity nobody could. That reading rests on assumptions that are rarely stated (Table 2). Two of them do worse than add noise: they make the decomposition report high confidence exactly where confidence is least warranted.

4.1. The floor moves when the problem changes

The Bayes risk is a floor set by the world *together with* the way the problem was posed: which features are measured,

Table 2. Conditions under which the two terms mean what their names suggest. The identity holds without any of them; only the interpretation needs them.

Assumption	What fails without it
Class contains the truth	residual absorbed by the noise scale; aleatoric term inflated (Section 3)
Converged, diverse posterior	otherwise disagreement measures the optimiser or the seed, not the evidence
Sample is representative	posterior contracts around the wrong answer; epistemic term falls while error does not (Section 5.3)
Test resembles training	both terms are extrapolated, and the aleatoric head extrapolates too (Hein et al., 2019; O’vadia et al., 2019)
Labels measure the target	aleatoric term measures the annotation process, not the world

how the target is defined, and how the labels were produced. Change any of those and the floor moves. For any covariate Z that could have been measured but was not, the law of total variance gives $\text{Var}(y | x) = \mathbb{E}[\text{Var}(y | x, Z) | x] + \text{Var}(\mathbb{E}[y | x, Z] | x)$. The second term is a variance, hence non-negative, so enriching the representation can only lower the Bayes error, with equality exactly when Z says nothing about y (Section A.6). There is no measurement that makes a problem intrinsically harder. What is filed as irreducible noise is in part missing sensors, coarsened features, an underspecified target (Aroyo & Welty, 2015; Plank, 2022) and protocol label noise (Northcutt et al., 2021), all of which are removable. Only outcomes not yet determined at prediction time resist every feature that could be added. “We have hit the noise floor” is therefore never a conclusion on its own; it is a conclusion about the current features, target and protocol, and it should prompt the question of whether any of the three can be improved.

4.2. Biased samples lower epistemic uncertainty

The failure modes above concern the model. This one concerns the sample, and the decomposition gives no hint of it. If the training sample is not a representative draw from the deployment distribution, the posterior does not notice. It still contracts as the sample grows, at the usual rate, but around the parameter that best fits the distribution that actually generated the data (Berk, 1966; Kleijn & van der Vaart, 2012). Epistemic uncertainty therefore falls as more data arrives, exactly as the theory says it should, while error on the deployment distribution does not fall with it. Collecting more data through the same pipeline removes the only signal a practitioner might have used to notice the problem, and it does so while looking like healthy convergence. Active learning inherits the defect: no acquisition function can reach a region the sampling frame excludes. Dataset bias of this kind is pervasive (Torralba & Efros, 2011), and its signature is a system accurate on average and

Table 3. E1. $\Delta = u_{\text{ale}} - u_{\star}$ against the measured CIFAR-10H floor. The gap is large in both directions and crosses zero between 176K and 701K parameters. Bootstrap 95% intervals are within ± 0.006 throughout.

model	params	acc	u_{epi}	Δ
linear	30.7K	0.403	0.0039	+1.609
small CNN	25.6K	0.815	0.0736	+0.473
r18/8	176K	0.918	0.0665	+0.065
r18/16	701K	0.942	0.0544	−0.043
r18/32	2.80M	0.956	0.0412	−0.085
r18/64	11.2M	0.960	0.0344	−0.100
r18/128	44.7M	0.964	0.0293	−0.106

confidently wrong on the under-represented group (Buo-lamwini & Gebru, 2018). Only external diagnostics see it, namely evaluation on a sample constructed by a different route and worst-group rather than average error (Sagawa et al., 2020; Koh et al., 2021). Section 5.3 measures it.

5. Evidence

Three experiments, numbered as in the pre-registration (Section C). E1 is the load-bearing one and was run first because it could falsify the position; E4 checks the mechanism of Proposition 3.1 where the truth is known; E3 concerns the sample rather than the model. A fourth, E2, was registered and not run, for reasons given in Section C. Protocol in Section B.

5.1. E1: the sign of the gap cannot be assumed, and it flips

Deep ensembles ($M = 5$) on CIFAR-10 across seven capacities, from a linear probe to a 44.7M-parameter ResNet, against the CIFAR-10H floor (Peterson et al., 2019): 511,000 judgements, 47–63 annotators per image, Miller–Madow corrected (plug-in bias would fabricate a trend of exactly our shape; Section B.2). Measured floor: mean 0.164 nats, 43.9% of images exactly zero. Training is plain, with no label smoothing, mixup or temperature scaling, all of which raise entropy; final train loss is 0.002 at the top, so the gap is not undertraining.

The gap is large in both directions. The position is that the sign and magnitude of Δ are not knowable without the floor, and Table 3 is that claim on real data. Across a single architecture family under a single recipe, Δ runs from +1.609 nats to −0.106, crossing zero at a capacity a practitioner has no way to locate from the model’s own outputs. Nothing in u_{ale} , in u_{epi} , or in accuracy says which side of the floor a given model sits on.

The crossing is a curve, not a point. Stratifying by measured floor changes the reading (Figure 1). On the 43.9%

of images where annotators were unanimous the floor is zero, so $\Delta = u_{\text{ale}} \geq 0$ *structurally*: nearly half the test set cannot go negative and drags the aggregate up. Reported per stratum, the capacity at which Δ changes sign falls monotonically as the floor rises: 549K, 222K, 82K and 26K parameters for the four non-unanimous strata, a $21\times$ range. *The more genuinely ambiguous the input, the smaller a model needs to be before it starts under-reporting that ambiguity.* On the most ambiguous stratum (floor 1.237 nats) the largest ensemble reports 0.976 nats *less* uncertainty than the annotators exhibit.

The epistemic term gives no warning. u_{epi} is non-monotone in capacity: it ranks the small CNN (81.5% accurate) above the ResNet (96.0%), never exceeds 0.074 nats while Δ traverses 2.6, and at the top is $28\times$ smaller than the error the aleatoric term makes on the hardest stratum. Ensembling does not help: u_{ale} is the mean of member entropies, untouched by M .

Annotator error. The CIFAR-10H floor contains annotator error as well as image ambiguity, and the two cannot be separated per image without a model of the annotators. This matters because annotator error inflates the floor, which pushes Δ negative, the direction of our finding: a model that is confidently right where an annotator mis-clicked is better than the annotator, not overconfident. By Section 4 the measured floor is the floor for the annotation protocol, and the position claims nothing beyond it; but a negative gap should not be read as a statement about the world unless it survives the confound. Table 4 tests whether it does. Modelling annotator error as symmetric mis-clicks at rate ε and deconvolving it from every image’s label distribution before recomputing the floor, the 43.9% unanimity rate bounds ε at about 1.6% on unambiguous images, and at that rate the aggregate gap moves by 0.004 nats. At $\varepsilon = 0.15$, nearly ten times the bound, the aggregate is still −0.057 and the ambiguous strata are below −0.4. Closing the aggregate gap needs a per-annotator error rate of order 20%, and no rate we tried closes it on the strata where the result lives. Baan et al. (2022) name this confound as open in their own setting; a Dawid–Skene deconvolution with per-annotator confusion matrices would tighten the bound and is the natural next step.

Resolution. One dissenting annotator in 51 gives an entropy of 0.097 nats, so that, not $1/k$, is the granularity of the floor near unanimity. The 43.9% of images measured at exactly zero have a true floor anywhere in $[0, 0.097)$, and the +0.027 we report on that stratum is below resolution and supports no claim about its sign. The same fact means the floor is underestimated on nearly half the test set, which biases the aggregate Δ upward by up to 0.04 nats; that is conservative for a negative-gap finding, and it is a further

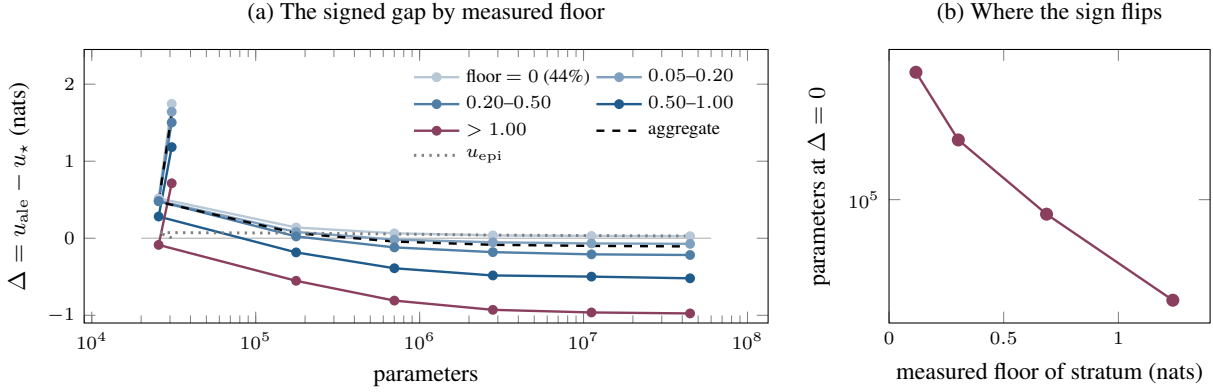


Figure 1. E1. (a) The gap between learned aleatoric uncertainty and the measured CIFAR-10H floor, by floor stratum, across the capacity ladder. The stratum with floor exactly zero cannot go negative; every other stratum does. The epistemic term (dotted) never leaves the neighbourhood of zero while Δ traverses 2.6 nats, and is non-monotone. (b) The capacity at which the sign flips falls monotonically as the floor rises: the more ambiguous the input, the smaller the model that already under-reports it.

Table 4. E1 sensitivity. Δ for the largest ensemble after deconvolving symmetric annotator error at rate ε from the CIFAR-10H labels. The unanimity rate bounds ε at ≈ 0.016 ; the finding survives ten times that.

ε	0.05-0.20	0.20-0.50	0.50-1.00	> 1.00	aggregate
0	-0.073	-0.217	-0.520	-0.976	-0.106
0.016	-0.067	-0.207	-0.510	-0.966	-0.102
0.05	-0.054	-0.185	-0.488	-0.943	-0.092
0.10	-0.033	-0.149	-0.452	-0.906	-0.076
0.15	-0.009	-0.107	-0.411	-0.861	-0.057
0.20	+0.027	-0.047	-0.353	-0.796	-0.030

Table 5. E4. The learned noise scale tracks $s^2 + b^2$, not s^2 . A misspecified linear head reports $11.3\times$ the true noise variance while its epistemic term is 0.0016 nats.

width	$\sigma^2_{\text{learned}}$	$s^2 + b^2$	$\times$ true	u_{epi}
0 (linear)	0.4507	0.4398	11.27	0.0016
2	0.0416	0.0409	1.04	0.0070
4	0.0408	0.0408	1.02	0.0081
16	0.0405	0.0411	1.01	0.0115
128	0.0377	0.0438	0.94	0.0384

reason to report per stratum.

5.2. E4: the noise scale absorbs the residual

Ensembles ($M = 5$) with heteroscedastic Gaussian heads on $y = \sin(x) + \varepsilon$, true $s^2 = 0.04$ known by construction, hidden width swept from 0 (linear) to 128. Because the truth is known, $b_\theta(x)$ is measurable and Proposition 3.1 is testable rather than merely illustrative.

Three results. The identity of Proposition 3.1 holds on trained networks: at width 0 the head reports 0.4507 against a predicted $s^2 + b^2 = 0.4398$. The second term of Equation (4) is present and the right size: a centred exponential noise of identical variance moves Δ from ≈ 0.00 to ≈ 0.40

nats, against the analytic $\text{KL}(\text{Exp} \parallel \mathcal{N}) = \frac{1}{2} \log(2\pi e) - 1 = 0.419$ that holds for every rate. And at width 128 the learned scale falls *below* the true noise, $\Delta = -0.069$, with no softmax present: the finite-sample deflation of Section 3. The table’s mean $\sigma^2 = 0.0377$ against $s^2 = 0.04$ would naively give $\frac{1}{2} \log(0.0377/0.04) = -0.030$; the difference is Jensen’s inequality, since Δ averages $\log \sigma^2$ over inputs and the fitted variance varies pointwise, and it implies a log-standard-deviation of about 0.4 in $\hat{\sigma}^2$ across the grid, the signature of a head that has fitted the sample rather than the noise. The β -NLL correction (Seitzer et al., 2022) barely moves the misspecified case ($11.27 \rightarrow 11.20$); it repairs an optimisation pathology, not a term the class cannot represent.

5.3. E3: biased frames lower epistemic uncertainty

Two groups with different true functions, deployment balanced, training frame biased with probability ρ of drawing group 0.

As ρ rises the epistemic term behaves correctly, reaching 1.891 nats at $\rho = 0.999$ (two minority examples). At $\rho = 1.000$, where the frame cannot reach the group, it *collapses* to 0.209 while worst-group error worsens $3.3\times$. Two minority examples to none makes the model more confident and more wrong: with no data every member extrapolates identically, and disagreement vanishes where it was needed (Hein et al., 2019). In the second sweep, with the group unreachable, $32\times$ more data buys a 7% reduction in worst-group error while reported uncertainty *falls*; the control with the minority merely rare buys $274\times$, with u_{epi} falling $157\times$. When the frame reaches the group, error and uncertainty fall together; when it cannot, error stalls and uncertainty keeps falling, and no term separates the two cases (Berk, 1966; Kleijn & van der Vaart, 2012).

5.4. What the three together establish

E1 shows that on a real benchmark the gap is large in both directions, that its sign depends on capacity and on the input in a way the model’s own outputs do not reveal, and that the flip is earliest where ambiguity is greatest. E4 shows the mechanism where the truth is known and both terms of Proposition 3.1 separate. E3 shows the sample causes a version of the failure that no term reports. In all three the epistemic term fails to warn; in two it moves the wrong way.

Limitations. None is a one-shot-outcome task, where the objection of Section 6 bites hardest. E1’s floor is CIFAR-10H’s annotation protocol, not the world’s, and the annotator-error analysis above bounds rather than removes that dependence. The resolution near unanimity is 0.097 nats, so the unanimous stratum supports no sign claim. All numbers are one recipe on one benchmark; label smoothing or mixup would raise u_{ale} and move the crossing, and establishing by how much is future work.

6. Alternative Views

Five positions oppose ours. The second is the strongest and we think it is substantially correct; we narrow the position rather than defend it. The fourth and fifth are both partly right, and both, followed through, end up asking for the measurement we are asking for.

1. “This is already known.” The model-relativity of the split is stated by Hüllermeier & Waegeman (2021) and Der Kiureghian & Ditlevsen (2009); Wimmer et al. (2023) question the measures; and Jiménez et al. (2025) have argued, in a position paper, that unaccounted bias inflates the aleatoric estimate and deflates the epistemic one. On this view the paper restates an argument that has already been made.

Response. The bias-contamination argument has been made, and we cite it as established rather than claim it. Three things are not in it. First, its direction: it is a one-sided story in which the aleatoric term is too large, and Section 3 shows that story is a population statement, with fitting the sample and the softmax’s sharpness both pushing the other way, so the sign of the error is open in both settings and the two mechanisms cancel. Second, a measurement: Jiménez et al. (2025) argue the contamination exists; E1 measures it against 511,000 human judgements and finds that it changes sign. Third, the practice, which we counted rather than asserted (Section D): in a random sample of 50 recent arXiv papers, of the 38 that report a learned aleatoric estimate only 5 compare it to any real external floor and 22 compare it to nothing, and half call the quantity *irreducible*, and that word is a claim. “Aleatoric uncertainty is model-relative” is a statement nobody has to act on; “the reported floor sat 0.98

nats below the measured one on the ambiguous stratum” goes in a table and moves a decision.

2. “Noise floors are unmeasurable in most real tasks.”

Repeated independent labels exist for image classification and some NLP annotation tasks. They do not exist for one-shot outcomes (did this loan default, did this patient survive), for time series where the label is realised once, or for measurements too expensive to replicate. A reporting requirement that most papers cannot satisfy is a bad requirement.

Response: we concede this, and it bounds the position. The claim is not that every project can supply a floor; it is that irreducibility requires evidence, and that three cases should be reported differently. Where repeated labels exist, measure the floor and report the signed gap. Where they do not, partial proxies usually do: inter-annotator agreement on a subsample, replicate measurements, test–retest reliability, or the residual entropy of a much larger model as a bound on what the current class is missing. A proxy that bounds the floor from one side is worth more than an assumption that bounds it from neither. Where neither exists, the correct report is that the estimate carries an error of unknown magnitude and sign, which is what it always was. That costs one sentence, and the sentence is true.

We take the residual force seriously. If few tasks admit any usable proxy, the position degrades from a protocol to a disclaimer. Our evidence that they are not few is confined to vision and tabular benchmarks and does not settle the question for, say, forecasting.

3. “Abandon the decomposition rather than repair it.”

If the aleatoric term needs an external measurement to be interpretable, the decomposition is doing little work. Conformal prediction gives distribution-free marginal coverage with no split at all (Vovk et al., 2005; Angelopoulos & Bates, 2023), and a practitioner who wants calibrated intervals should use it and stop there.

Response. Conformal prediction answers a different question: how wide an interval must be for a stated coverage, not *why*, and so not whether to buy labels, build a sensor or change the architecture. That question is the one the split was introduced to answer. Our position is compatible with conformal coverage and concerns what may be claimed when the split is reported. Abandonment carries the heavier burden: it must make the acquire-data versus improve-features distinction some other way, and we know of no method that does.

4. “Inflated aleatoric estimates are still useful.” For selective prediction, an upper bound on label ambiguity is exactly what is wanted: a model that abstains where its

own noise estimate is large will abstain where it is wrong, whether the cause is genuine ambiguity or a class that cannot fit the region (Geifman & El-Yaniv, 2017). Tightening the estimate against a floor would, on this view, make the abstention rule worse.

Response. We agree with the premise and it is the cleanest illustration of what the position does and does not target: abstention uses the quantity, not the interpretation. Nothing here asks anyone to stop computing the aleatoric term or to stop abstaining on it. What it asks is that the sentence “this uncertainty is irreducible” not be attached to it without evidence, because that sentence, unlike the abstention rule, terminates work that would otherwise continue. An upper bound is a fine trigger and a poor conclusion.

But the premise fails where it matters. Proposition 3.2 shows the estimate is not an upper bound in classification, and E1 finds it *below* the floor at deployment capacities, so a classifier abstaining on it under-abstains exactly where annotators disagree, the inputs where abstention was the point. Whether an aleatoric estimate is safe to abstain on is precisely the question a floor answers.

5. “Calibration already covers this.” If the complaint about classification is that models are overconfident relative to human agreement, that is miscalibration, and temperature scaling and its successors address it (Guo et al., 2017). Baan et al. (2022) go further and propose instance-level calibration against the full distribution of human judgements rather than the majority vote. On this view the floor adds nothing that calibration against the same repeated labels does not already give.

Response. Baan et al. (2022) is the closest precedent for Section 7, and we regard their measures as the calibration form of the same audit. The difference is what is being asked. Calibration asks whether the predictive distribution matches the human one; the position asks what may be claimed about the aleatoric term when it does not, and specifically whether the residual belongs to the world or to the model. Proposition 3.2 separates the two mechanisms per input; a calibration statistic, instance-level or not, reports their sum, and a model can be calibrated in aggregate while over-reporting where the class cannot fit and under-reporting where it is overconfident. And calibrating against repeated labels *is* measuring the floor: an objection that recommends collecting the data we ask for is a route to satisfying the position, not an objection to it.

7. Call to Action

The position implies a reporting protocol, not a new method; it costs one held-out subset and a paragraph.

For anyone reporting an aleatoric estimate. Report it against a floor, or report that you cannot. Concretely: collect $k \geq 2$ independent labels on a held-out subset (a few hundred items is usually enough to separate the two quantities), estimate the floor u_* as the bias-corrected conditional entropy of the label distribution under the annotation protocol, and report the triple $(u_{\text{ale}}, u_*, \Delta = u_{\text{ale}} - u_*)$ rather than u_{ale} alone. *Report Δ with its sign, and do not take $|\Delta|$ small as reassurance:* by Proposition 3.2 the two mechanisms that move it have opposite signs and cancel, so a small gap is consistent with both being large. Where repeated labels are impossible, name the proxy used or state that the estimate carries an error of unknown magnitude and unknown sign. Never write *irreducible* without one of the two.

Read a floor that has been reached as a question, not an answer. The floor is set by the world together with the features, target definition and labelling protocol (Section 4); reaching it licenses the question of whether any of the three can be improved, and only outcomes not yet determined at prediction time resist every answer.

Stop ranking architectures by uncertainty magnitude. A more restrictive class admits fewer good fits, so its members agree more and it reports *less* epistemic uncertainty; Section 5.1 finds the term non-monotone in capacity. Rank by proper scoring rules against outcomes that happened (Brier, 1950; Gneiting & Raftery, 2007), and use the floor comparison for the architecture question.

For benchmark and dataset builders. Release repeated labels on the test split; CIFAR-10H (Peterson et al., 2019) and the ImageNet relabellings (Beyer et al., 2020; Northcutt et al., 2021) show the cost is modest. A benchmark without them cannot distinguish a model that has solved the task from one that has exhausted its class, and the two call for opposite responses.

For reviewers, and for method development. Treat “irreducible” as a claim requiring support, as a generalisation bound requires a statement of whether it is non-vacuous at the sample size in use. Ask what a new aleatoric estimator was validated against; the usual answer is a synthetic problem with noise known by construction, the one setting in which the failure cannot occur. And treat Δ as what it is, a supervised signal for misspecification that is currently unused: ensembling across architectures rather than seeds makes inadequacy partly visible (Wenzel et al., 2020; Fort et al., 2019), and a discrepancy term in the style of Kennedy & O’Hagan (2001) would give the residual somewhere to go other than the noise scale.

8. Conclusion

The uncertainty decomposition is exact, cheap, and computable without labels, and it is silent on the question most often asked of it. Approximation error has no term in it, by construction, and is reported instead as noise, under the name of a quantity supposed to be irreducible. The remedy is not a new estimator but a measurement the field already knows how to make and almost never makes. Report the floor, report the signed gap, and reserve the word *irreducible* for the cases that have earned it.

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A. Derivations

A.1. Proof of Proposition 3.1

Fix x and write $m = m(x)$, $s^2 = s^2(x)$, $\mu_\theta = \mu_\theta(x)$, $b_\theta = \mu_\theta - m$. The member’s conditional negative log-likelihood is

$$\begin{aligned} \mathbb{E}[-\log \mathcal{N}(y; \mu_\theta, \sigma_\theta^2) | x] \\ = \frac{1}{2} \log(2\pi\sigma_\theta^2) + \frac{\mathbb{E}[(y - \mu_\theta)^2 | x]}{2\sigma_\theta^2}. \end{aligned}$$

Differentiating in σ_θ^2 and setting to zero gives $\sigma_\theta^2 = \mathbb{E}[(y - \mu_\theta)^2 | x]$. Expanding about m and using $\mathbb{E}[y - m | x] = 0$,

$$\sigma_\theta^2(x) = s^2(x) + b_\theta(x)^2. \quad (7)$$

The differential entropy of $\mathcal{N}(\mu, \sigma^2)$ is $\frac{1}{2} \log(2\pi e \sigma^2)$, so $u_{\text{ale}}(x) = \frac{1}{2} \mathbb{E}_\theta \log(2\pi e(s^2 + b_\theta^2))$. For any law p with mean m and variance s^2 ,

$$\text{KL}(p \| \mathcal{N}(m, s^2)) = -\mathbb{H}[p] + \frac{1}{2} \log(2\pi e s^2), \quad (8)$$

since $\int p \log \mathcal{N}(y; m, s^2) dy = -\frac{1}{2} \log(2\pi s^2) - \frac{1}{2}$. Subtracting $u_\star(x) = \mathbb{H}[p(y | x)]$ and splitting the logarithm,

$$\Delta(x) = \frac{1}{2} \mathbb{E}_\theta \log\left(1 + \frac{b_\theta^2}{s^2}\right) + \text{KL}(p(y | x) \| \mathcal{N}(m, s^2)),$$

which is Equation (4). Both terms are non-negative, the first because $b_\theta^2 \geq 0$ and the second because it is a Kullback–Leibler divergence, and each vanishes exactly under the stated condition. $\square$

Two remarks. The Gaussianity assumed is that of the *head*, not of the world; a non-Gaussian truth enlarges the second term rather than invalidating the result. And the optimum in Equation (7) is taken over the true conditional law, which is what makes the conclusion a population statement; see Section A.4.

A.2. Proof of Proposition 3.2

Write $p = p(\cdot | x)$ and $p_\theta = p(\cdot | x, \theta)$. By definition of the cross entropy, $\text{CE}(p, p_\theta) = \mathbb{H}[p] + \text{KL}(p \| p_\theta)$, and by construction $C_\theta = \text{CE}(p, p_\theta) - \mathbb{H}[p_\theta]$. Eliminating CE,

$$\mathbb{H}[p_\theta] - \mathbb{H}[p] = \text{KL}(p \| p_\theta) - C_\theta.$$

Averaging over θ gives Equation (6). $\square$

$C_\theta = \langle p - p_\theta, -\log p_\theta \rangle$ is positive when p_θ concentrates its mass where its own surprisal is low relative to the truth, which is what it means to be sharper than the truth. It is exactly zero when p_θ is uniform, and negative when p_θ is more diffuse than p , in which case both terms inflate the reported floor.

A.3. The residual identity

Moment-matching the ensemble’s predictive mixture, $u_{\text{epi}}(x) \approx \frac{1}{2} \log(1 + \text{Var}_\theta(\mu_\theta)/\mathbb{E}_\theta \sigma_\theta^2)$. Using $\mathbb{E}_\theta b_\theta^2 = (\mathbb{E}_\theta \mu_\theta - m)^2 + \text{Var}_\theta(\mu_\theta)$ and expanding both logarithms to first order in the regime where the errors are small relative to the noise, the $\text{Var}_\theta(\mu_\theta)$ contributions cancel and

$$\begin{aligned} \Delta(x) - u_{\text{epi}}(x) &= \frac{(\mathbb{E}_\theta \mu_\theta(x) - m(x))^2}{2s^2(x)} \\ &\quad + \text{KL}(p \| \mathcal{N}) + O(\cdot), \end{aligned}$$

which is Equation (5). The surviving term is the squared *bias* of the ensemble: the error common to every member, which no amount of disagreement between members can reveal.

A.4. Finite samples break the one-sided guarantee

Equation (7) holds at the optimum over the true conditional law. A trained network optimises the empirical objective, and for an overparameterised head the training residuals are smaller than the true noise, so $\hat{\sigma}_\theta^2 < s^2 + b_\theta^2$ and possibly $\hat{\sigma}_\theta^2 < s^2$. Section 5.2 observes exactly this at the widest model in the sweep: $\hat{\sigma}^2 = 0.0377$ against a true $s^2 = 0.04$, giving $\Delta = -0.069$ in a regression problem containing no softmax. The unified statement in Section 3, that misspecification inflates the reported floor while fitting the sample deflates it, is the reading that survives in both settings.

A.5. Non-negativity of the epistemic term

$\mathbb{H}[\cdot]$ is concave, so by Jensen’s inequality $\mathbb{H}[\mathbb{E}_\theta p_\theta] \geq \mathbb{E}_\theta \mathbb{H}[p_\theta]$: averaging the distributions first captures both each member’s spread and the spread between members, while taking entropies first captures only the former. The difference is the disagreement, and it cannot be negative.

A.6. The floor moves with the information set

For any covariate Z that could have been measured but was not, the law of total variance gives

$$\text{Var}(y | x) = \mathbb{E}[\text{Var}(y | x, Z) | x] + \text{Var}(\mathbb{E}[y | x, Z] | x).$$

The second term is a variance, hence non-negative, and vanishes exactly when Z carries no information about y given x . So enriching the representation can only lower the Bayes error: *there is no measurement that makes a problem intrinsically harder*. What is filed as irreducible noise at resolution x is, in part, signal that a richer feature set would have recovered.

B. Experimental Detail

All experiments ran on a single NVIDIA A100-SXM4-80GB. The complete E1 sweep, 35 training runs, took ap-

proximately 3.3 hours wall-clock; E3 and E4 together took under an hour. Code and all raw outputs accompany the submission.

B.1. E1: CIFAR-10 against CIFAR-10H

Capacity ladder. One architecture family with a width knob, so the abscissa is parameter count rather than a mixture of unrelated designs: a linear probe on raw pixels (30.7K parameters), a two-layer CNN (25.6K), and a CIFAR ResNet-18 (3×3 stem, no max-pool) at base widths 8, 16, 32, 64, 128 (176K to 44.7M).

Recipe. SGD with Nesterov momentum 0.9, weight decay 5×10^{-4} , batch size 128, 100 epochs, one-cycle cosine schedule (pct_start 0.15, initial division 10, final division 100) with peak learning rate 0.01 for the linear probe, 0.05 for the CNN and 0.1 for the ResNets. Augmentation is reflect-padded random 32×32 crops and horizontal flips, nothing more. bfloat16 autocast, channels_last. $M = 5$ members per capacity, differing by seed.

We use no label smoothing, no mixup and no temperature scaling. All three raise predictive entropy and improve calibration, and any of them could mask the effect being measured; a study that adopted a modern default recipe and found no gap would be measuring its own regulariser. Establishing whether label smoothing closes the gap is worth doing separately, and we flag it as future work rather than folding it in silently.

Convergence. Final training loss is 1.817 (linear), 0.675 (CNN), 0.183, 0.043, 0.009, 0.003 and 0.002 across the ResNet ladder. By Proposition 3.1 the gap reports total excess risk, so an undertrained model would inflate Δ through the optimisation term; at the capacities where we report a negative gap that term is negligible.

Index alignment. CIFAR-10H rows correspond to the CIFAR-10 test set in its original order. We verify rather than assume this: the annotator majority vote agrees with the test label on 99.21% of images. A misaligned join would show agreement near 10%, and the evaluation aborts if agreement falls below 90%.

B.2. Estimating the floor

CIFAR-10H provides 511,000 judgements over the 10,000 test images, 47–63 annotators each (mean 51.1); 43.9% of images are unanimous and 90% have support of at most three classes. Estimator choice is not a detail here, because the predicted negative gap and the plug-in estimator’s bias have the same sign.

A parametric bootstrap, taking the empirical per-image distribution as truth and redrawing that image’s actual k , gives

Table 6. Measured floor under each estimator (nats), real CIFAR-10H counts.

estimator	mean	median	p90
plug-in	0.1545	0.0950	0.4643
Miller–Madow	0.1637	0.1046	0.4890
Grassberger	0.1813	0.1214	0.5132
Krichevsky–Trofimov ($\alpha = \frac{1}{2}$)	0.5772	0.5198	0.8262
Laplace ($\alpha = 1$)	0.8444	0.7908	1.0567

the bias by stratum:

Table 7. Estimator bias by true floor (nats). Plug-in bias grows $3.7\times$ across strata and so fabricates a negative trend of the shape we claim; Miller–Madow removes level and trend alike.

true floor	plug-in	Miller–Madow	Grassberger
$\approx$ unanimous	+0.0000	+0.0000	+0.0096
0.05–0.20	−0.0128	−0.0052	+0.0092
0.20–0.50	−0.0220	−0.0066	+0.0089
0.50–1.00	−0.0304	−0.0076	+0.0086
> 1.00	−0.0468	−0.0078	+0.0100

We therefore use Miller–Madow throughout. Grassberger is flat and *positive*, so it errs against a negative-gap finding and serves as the robustness check. Add- α smoothing must not be used: at $K = 10$, $k \approx 51$ it inflates the floor by 0.41–0.68 nats by spreading pseudo-counts over roughly nine empty cells on near-unanimous images, which would fabricate a large positive gap. A simulation with synthetic label distributions predicted +0.413 and +0.678 before the data was obtained, so this is a property of the regime rather than of one dataset. Since Section 7 asks practitioners to measure floors, and add- α is a common default, the warning belongs with the recommendation.

Minimum detectable $|\Delta|$ at two standard errors is 0.003, 0.006, 0.009 and 0.020 nats for the four non-unanimous strata, 5 to 60 times the resolution the reported effects require. Per-image standard deviation is 0.076 nats, however, against effects of order 0.1, so we make no per-image claims and report stratum means throughout.

Resolution near unanimity. A single dissenting annotator among 51 gives $\mathbb{H}[(50/51, 1/51)] = 0.097$ nats, so the floor is quantised at roughly 0.1 nats near zero, not at $1/k$. Both plug-in and Miller–Madow return exactly 0 on a unanimous image, so the 43.9% of images at zero have a true floor in $[0, 0.097)$ and the aggregate Δ is biased upward by up to 0.04 nats. This is conservative for a negative-gap finding and is stated in Section 5.1.

Annotator error. Table 4 models annotator error as symmetric mis-clicks at rate ε , $p_{\text{obs}} = (1 - \varepsilon)p + \varepsilon/K$, and deconvolves it from every image’s label distribution, clipping at zero and renormalising, before recomputing the floor.

Strata are fixed by the original floor so the comparison is like for like. The unanimity rate bounds ε : on an unambiguous image $P(\text{unanimous}) = (1 - \varepsilon)^{51}$, and 43.9% unanimity gives $\varepsilon \lesssim 0.016$. Symmetric error is the least favourable model for us, since it spreads mass onto all nine wrong classes and so inflates entropy maximally per unit of error; a per-annotator confusion model in the style of Dawid and Skene would give a tighter bound and is left as future work.

B.3. E4: sine toy

$y = \sin(x) + \varepsilon$, $x \sim \mathcal{U}(-4, 4)$, $n = 512$, true $s^2 = 0.04$. Two-hidden-layer tanh networks emitting μ and $\log \sigma^2$ (clamped to $[-12, 6]$), width 0 (linear) through 128; $M = 5$; Adam at 3×10^{-3} with cosine decay for 6000 full-batch iterations. Evaluation on a 2000-point grid. The exponential arm draws ε from a centred exponential of identical variance, for which $\text{KL}(\text{Exp} \parallel \mathcal{N}) = \frac{1}{2} \log(2\pi e) - 1 = 0.4189$ nats for *every* rate, a fixed analytic target for the second term of Equation (4). The β -NLL arm uses $\beta = 0.5$ (Seitzer et al., 2022).

B.4. E3: biased sampling frames

Two groups on $x \sim \mathcal{U}(-2, 2)$ with $s^2 = 0.04$: group 0 follows $\sin(3x)$, group 1 follows $-\sin(3x) + \frac{1}{2}x$. The training frame draws group 0 with probability ρ ; deployment is balanced. Width-64 two-layer tanh heteroscedastic networks, $M = 5$, Adam at 3×10^{-3} with cosine decay for 4000 iterations, evaluated on a 500-point grid per group.

Our first design for the second sweep held $\rho = 0.95$, where the minority still receives 5% of n and is comfortably learnable; error and epistemic uncertainty fell together and there was no dissociation. That is a real boundary on the claim rather than a failed experiment, so we report it as the control alongside the $\rho = 1.00$ sweep the claim is actually about.

C. Pre-registered Predictions

Predictions for four experiments were written down before any model was trained; three were run and the register accompanies the submission. The fourth, E2, a UCI regression study ranking architectures by mean epistemic uncertainty against held-out log-likelihood, was not run: E1 already demonstrates its claim on real data, with the epistemic term non-monotone in capacity, and we preferred to report that than to add a confirmatory synthetic result. We keep the original numbering so the register and the paper agree.

Direction was correct on every stratum and on the aggregate; magnitudes were consistently 25–30% conservative, meaning the models are more overconfident than we assumed. Two predictions were wrong in ways that changed the paper, and we record them rather than quietly repairing them.

Table 8. E1 at the 11.2M-parameter ResNet: predicted versus measured Δ (nats).

stratum	predicted	measured
$\approx$ unanimous	+0.03	+0.0304
0.05–0.20	−0.05	−0.0671
0.20–0.50	−0.15	−0.2089
0.50–1.00	−0.39	−0.4980
> 1.00	−0.74	−0.9631
aggregate	−0.07	−0.0995

Proposition 1 was expected to bound the gap below by zero in practice. It does not: finite-sample shrinkage of the variance head drives Δ negative at high capacity in regression (Section A.4). This is what replaced the original one-sided regression story with the unified statement in Section 3, and it strengthened the argument rather than weakening it.

E3’s second sweep was first designed at $\rho = 0.95$, a regime where the minority is still reachable and the predicted dissociation does not occur. The corrected design and the original as control are described in Section B.

D. Reporting Survey

The claim in Section 1 and Section 6 that learned aleatoric estimates are reported without comparison to any measured floor deserves a count rather than an assertion. This is a pilot count, with its protocol fixed before any paper was coded; the full protocol, sample, every verdict with its quoted evidence, and every audit override with its reason accompany the submission.

Frame and sample. arXiv papers whose abstract contains “aleatoric”, listed in cs.LG, stat.ML or cs.CV, submitted 2025-09-01 through 2026-08-31, the twelve complete months before the survey date: 201 papers. A uniform random sample of 50 (fixed seed) was coded.

Coding. Three questions per paper, from the arXiv full text (one paper from its abstract only). Q1: does the paper *report* a learned or estimated aleatoric quantity on some task, as opposed to only discussing the concept? Q2, if so: is that quantity compared to any external measurement or proxy of label noise, coded *real* (repeated labels, multiple annotators, human disagreement, replicate measurements, or an external estimate of the achievable floor on real data), *synthetic* (only a noise level the authors set by construction), or *none*? Q3: does the paper describe aleatoric uncertainty as irreducible or as uncertainty that cannot be reduced? A fixed prompt was run once per paper by an automated reader, which returned each verdict with the passage it relied on. Every quoted passage was then audited by hand against the coding rules, and a verdict was overridden only where the

quote plainly failed its rule; the audit touched 20 papers and every override is recorded with its reason. Both the raw and the audited counts are reported.

Table 9. Reporting survey, $n = 50$. Shares of Q2 are among papers with Q1 = yes. Wilson 95% intervals.

	raw	audited	95% CI
reports a learned aleatoric estimate (Q1)	40/50	38/50	
compared to a real external floor	7/40	5/38	[0.06, 0.27]
compared to a synthetic level only	13/40	11/38	
compared to nothing	20/40	22/38	[0.42, 0.72]
calls it irreducible (Q3)	43/50	25/50	[0.37, 0.63]

Result. Of the 38 sampled papers that report a learned aleatoric estimate, 5 compare it to a real external floor (13construction, and 22 to nothing at all (58Half of the sample calls the quantity irreducible. So “almost nobody” would overstate it: about one paper in eight makes the comparison, and those papers are the practice Section 7 recommends, already in use. Two of them train or evaluate a learned aleatoric head against multi-annotator label distributions, one estimates annotator ambiguity directly, and one uses a betting market as an external ceiling on achievable forecast quality. One of the five was coded from its abstract only and is uncertain; excluding it gives 4/38. The majority, and the word, are the premise of the position.

What the audit changed. The automated reader over-called “irreducible”: it answered yes for 43 papers, but in 18 of those the quoted passage said “inherent”, “intrinsic” or nothing of the kind, and the audit reversed them. It also coded two qualitative descriptions of hard inputs as real-floor comparisons and two non-estimates as estimates. The audited real-floor count is therefore lower than the raw one, not higher; the correction runs against the paper’s convenience.

Limitations. A pilot of 50 from a frame of 201; one automated pass with a human audit of the quotes rather than of the full papers; an arXiv frame, so unrefereed work is included; and a keyword frame, so papers reporting the quantity under another name are missed. The camera-ready survey should code the full frame, add a second independent human coder on a subsample, and report agreement.

E. A Working Checklist

Questions worth asking of one’s own results before asking them of anyone else’s.

- Which term am I trying to reduce: approximation, estimation or optimisation? Have I confirmed the diagnosis before choosing the remedy?

- Do I have any estimate of the noise floor for this task, and where does my model’s aleatoric term sit relative to it: above, or *below*?
- If I have hit the floor, have I asked whether better features, a sharper target definition or a better labelling protocol would lower it?
- Am I reading a high aleatoric estimate as irreducible, when it may be telling me my features are inadequate?
- Am I ranking architectures by uncertainty magnitude anywhere?
- Is my out-of-distribution detection resting on epistemic uncertainty rather than on an explicit density or distance model?
- Is my training sample a representative draw from the distribution I will deploy into? No uncertainty term will tell me if it is not.
- Is the reported improvement larger than seed-to-seed and split-to-split variance?
- How many times has this evaluation set influenced a decision?
- Does the deployment distribution match the evaluation distribution, and which groups would reveal it if it did not?
- If I am citing a generalisation bound, is it non-vacuous at my actual sample size and model scale?